

COMP 3311: Database Management Systems

MongoDB Database Creation and MQL Queries Assignment

Assigned: July 16, 2026

Value: 10% of course grade

Due: 11:00 p.m., July 25, 2026

For this assignment you are to submit an SQL script file containing SQL statements that create MongoDB collections from a relational database for the movie streaming management system and an MQL query file that executes some queries on these collections.

IMPORTANT REMINDER

This is an individual assignment. The submitted SQL script file and MQL query file should be *your own work*. While you may discuss general assignment issues with other students, you are not allowed to collaborate with any person, AI chatbot or AI agent to arrive at a solution, to share solutions or to copy someone else's solution. Copying, sharing and collaborating will be severely penalized. All those involved in a copying/sharing/collaborating case will automatically receive a grade of 0 and may be reported for further disciplinary action.

PART 1: MONGODB DATABASE CREATION

In the SQL script file **mqlCreatedb.sql**, construct, using the Oracle SQL/JSON functions presented in the course, SQL statements that create JSON collections for a MongoDB database with the structure indicated below from a relational database whose relation schemas are those specified in the **msmsSampledb.sql** script file. In all created collections, a field whose value is null should be **absent** in objects/arrays of the collection.

The following is an example of an object in the **moviePersons** collection.

```
{personId: 104, name: "Adam McKay", biography: "Adam McKay is an American actor, film director, producer and screenwriter.",
gender: "M", birthdate: "17-APR-1968", deathdate: "12-FEB-2026",
  appearsIn: [11, 36, ...], /* an array of movie ids */
  directs: [11, 25, 44, ...], /* an array of movie ids */
  awardWin: [{movieId: 11, awardName: "Best Actor"}, {movieId: 25, awardName: "Best Director"}, ...], /* an array of objects */
}
```

The following is an example of an object in the **movies** collection.

```
{movieId: 25, title: "One Battle After Another", synopsis: "When their enemy resurfaces after 16 years, a group of ex-revolutionaries
reunite to rescue the daughter of one of their own.", releaseYear: "2025", runningTime: 161, mpaaRating: "R", imdbRating: 7.6,
award: "Best Picture",
  genres: ["dark comedy", "drama", ...], /* an array of genres */
  cast: [{personId: 101, role: "Bob"}, {personId: 105, role: "Col. Steven J. Lockjaw"}, ...], /* an array of objects */
  directors: [104, ...], /* an array of person ids */
  reviews: [{username: "adamau", title: "Perfect", rating: 9, reviewText: "I cannot think of one reason I would not give this 10/10
except out of principle.", reviewDate: "05-JAN-2026"}, ...] /* an array of objects */
}
```

The following is an example of an object in the **reelfliesMembers** collection.

```
{username: "adamau", pseudonym: "hitchcoc", firstName: "Adam", lastName: "Au", occupation: "student", email: "adamau@nomail.com",
gender: "M", birthdate: "14-FEB-2007", phoneNumber: "93467812", educationLevel: "secondary", cardNumber: "6223948690316102",
cardholdername: "Adam Au", cardType: "UnionPay", security code: "187", expiryMonth: "08", expiryYear: "2027",
  watchlist: [11, ...], /* an array of movie ids */
  streamingHistory: [{movieId: 25, streamingDate: "01-JAN-2026"}, ...] /* an array of objects */
}
```

Note: You are required to use the SQL script file **mqlCreatedb.sql** for Part 1.

IMPORTANT NOTES

1. You must use the Oracle SQL/JSON functions presented in the course in your SQL statements that create the collections described above. No marks will be given if these functions are not used even if your SQL statements correctly create the collections.
2. The collection and field names in the collections that your SQL statements create should be exactly the same as those shown in the example objects above as case matters in MongoDB.

PART 2: MQL QUERIES

In the text file **mqlQueries.txt**, construct MQL queries in the indicated places that use the collections specified in **Part 1** to retrieve the required information as specified by the query statements.

Note: *You are required to use the query text file **mqlQueries.txt** for Part 2.*

IMPORTANT NOTES

1. You are allowed to use only MQL constructs that have been discussed in the course. Moreover, you are not allowed to use nested pipelines (e.g., pipeline lookup) or JavaScript code in your queries. No marks will be given for a query if these requirements are not followed.
2. The collection and field names in your MQL queries should be exactly the same as those shown in the examples in Part 1 as case matters in MQL queries.

Example Query Results

Below are example query results for each query illustrating how the query result should be presented.

Query 1: {title: 'Sunset Boulevard', genres: ['dark romance', 'drama', 'film-noir']}

Query 2: {name: 'Mel Gibson', role: 'William Wallace', title: 'Braveheart', releaseYear: '1995'}

Query 3: {name: 'Ralph Fiennes', movies: [{title: 'The Hurt Locker', releaseYear: '2008'},
{title: 'The English Patient', releaseYear: '1996'},
{title: 'Schindler's List', releaseYear: '1993'}]}

Query 4: {title: 'Hara-Kiri', imdbRating: 8.6, reelflicsRating: 9}

Query 5: {title: 'It Happened One Night', releaseYear: '1934', actor: 'Clark Gable', actress: 'Claudette Colbert',
directors: 'Frank Capra'}

Query 6: {title: 'Titanic', mpaaRating: 'PG-13', imdbRating: 7.8, reelflicsRating: 6.3, numStreamed: 1}

ASSIGNMENT FILES

All files required for this assignment are in the zipped folder **MongoDB Database Creation and MQL Queries Files** which can be downloaded from the entry for this assignment on the Canvas *Assignments* page.

WHAT TO SUBMIT

Submit your completed SQL script file **mqlCreatedb.sql** and MQL query file **mqlQueries.txt**.

Important Notes

1. The file **mqlCreatedb.sql** should not contain any statements that insert data; it should contain only the SQL statements for creating the collections.
2. Your student id and name should be included in the indicated place in each file.
3. We will not grade your submission if
 - your **mqlCreatedb.sql** script file violates any of the requirements stated in **Part 1**,
 - you create the script file **mqlCreatedb.sql** in other formats (i.e., doc, docx, rtf, pdf, txt, etc.),
 - you create the query file **mqlQueries.txt** in other formats (i.e., doc, docx, rtf, pdf, sql, etc.) or
 - you do not put your student id and name in each file correctly.

Note: *Your **mqlCreatedb.sql** script file will be run as a script using Oracle SQL Developer. If the script file cannot be executed, it will not be graded, and you will receive 0 for Part 1 of this assignment.*

PLEASE NOTE CAREFULLY

You may not share your SQL script file or your MQL query file with other students in the course. Sharing your script/query files will be considered collaboration and will be penalized as stated above.

HOW TO SUBMIT

1. Place your SQL script file **mqlCreatedb.sql** and your MQL query file **mqlQueries.txt** in a folder named **mqlXXXXXXXX** where **XXXXXXXX** is your student id.
2. Compress (zip) the folder **mqlXXXXXXXX**.
3. **By 11:00 p.m. on Saturday, July 25**, upload your **mqlXXXXXXXX.zip** file to the *MongoDB Database Creation and MQL Queries* assignment on Canvas.

It is your responsibility to ensure that your *mqlXXXXXXXX.zip* file is correctly uploaded to Canvas. Under no circumstances will late submission or submission by email be accepted. Moreover, any submission to the assignment's comments section will be ignored.

GRADING

<u>Item</u>	<u>Value</u>
Part 1: MongoDB database creation Grading will be based on correct generation of the collections as specified in this assignment.	~30%
Part 2: MQL queries Grading of each query will be as follows: 10% query raises error 25% no query result 50% query result contains none or only some of the expected results 75% query result contains all expected results plus additional incorrect results 90% query result contains only expected results, but there are duplicate results and/or not all expected fields are present and/or results are not ordered or formatted as specified	~70%

CLARIFICATION AND AMENDMENT OF ASSIGNMENT REQUIREMENTS

Clarification questions about the assignment should be submitted to the teaching team by email at 3311rep@cse.ust.hk. In addition to being answered individually, if appropriate, a submitted question and its reply will be posted in the *MongoDB Database Creation and MQL Queries Q&A* document on the Canvas page for this assignment. ***This document should be consulted on a regular basis for further clarification and amendment of assignment requirements.*** Any requirements added or amended in this document become part of the assignment requirements.